

Practical Work 3: MPI File Transfer

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1 Introduction

This practical extends the TCP file transfer system by using the Message Passing Interface (MPI). Instead of sockets, processes communicate via MPI point-to-point operations. The goal is to implement a simple file transfer system using two MPI processes.

2 Choice of MPI Implementation

I chose to use a standard MPI implementation such as MPICH or OpenMPI because they are widely available, portable and compatible with the `mpicc` and `mpirun` tools provided in the course environment.

3 Design of the MPI Service

The system uses exactly two processes:

- Rank 0: acts as the client. It reads the input file and sends data chunks to rank 1.
- Rank 1: acts as the server. It receives the chunks and writes them to the output file.

Each chunk transfer consists of two messages: one for the chunk length and one for the data itself. A chunk length of zero is used as an End-of-File marker.

4 System Organization

The practical is organized in a single C program:

- `mpi_file_transfer.c`: contains both client and server logic, selected by the MPI rank.

The program is compiled with `mpicc` and executed using `mpirun -np 2`.

5 Implementation Snippet

Example code from the sender (rank 0):

```
while (1) {
    int n = (int)fread(buffer, 1, CHUNK_SIZE, in);
    MPI_Send(&n, 1, MPI_INT, 1, TAG_CHUNK_LEN, MPLCOMM_WORLD);
    if (n == 0) break;
    MPI_Send(buffer, n, MPI_UNSIGNED_CHAR, 1, TAG_CHUNK,
             MPLCOMM_WORLD);
}
```

6 Group Work

This practical was completed individually.

7 Conclusion

By using MPI, the file transfer system no longer depends on TCP sockets. The communication is handled by MPI sends and receives, which simplifies some aspects of the networking code and makes it easier to scale to more processes in the future.